

Customer Complaint Triage and Escalation Modeling

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I. Introduction

The strategic management of consumer grievances is a critical imperative for financial institutions, where effective and timely resolution directly influences customer retention and mitigates severe regulatory risk. As the usage and penetration of Artificial Intelligence (AI) and Machine Learning (ML) expands in the financial domain, it is vital to move from black-box models towards regulated and explainable usage of AI. Having prior industry experience of 2 years in the fintech lending industry, the importance of integrating explainable AI methodology to solve business problems became evident to me. One of the most faced problems in service-based industry is that of customer churn. Additionally, customer satisfaction is prioritized in this sector. But a significant amount of time is spent on the analysis and processing of complaints which may lead to consumer dissatisfaction. The goal of this project is to develop a framework using the techniques of Natural Language Processing to analyze customer sentiment and layer it with a Hidden Markov Model to understand the escalation of complaints and tracking customer dissatisfaction states. This notion brought me to review existing literature on these topics and narrow down the dataset for my analysis. The Consumer Financial Protection Bureau (CFPB) provides an extensive public repository with millions of consumer complaints aggregated with time. This dataset is best suited for my analysis and project as it has the customer complaint narrative associated with the complaints. However, this dataset is inherently anonymized to protect personally identifiable information. The removal of unique customer identifiers presents a profound methodological barrier to traditional longitudinal journey mapping and churn prediction. Without the ability to track a single user over time, organizations are traditionally forced to rely on static snapshots of dissatisfaction, treating each complaint as an isolated event rather than a node in a continuous behavioral sequence. My project addresses this fundamental gap by proposing an "Intelligent Triage" framework that combines natural language processing (NLP) with a simulation-based Hidden Markov Model (HMM) to predict complaint resolution complexity.

The theoretical foundation of this work is built upon the intersection of sentiment analysis, topic modeling, and sequential behavioral prediction. Peng et al. (2024) established that lexicon-derived sentiment scores are not merely descriptive but carry statistically significant predictive signals for customer churn behavior. However, applying sentiment analysis to financial domains requires careful consideration of the text's inherent register. Kim et al. (2024) noted that standard sentiment tools often underperform when applied to the "unemotional prose" of formal financial documentation. Consequently, this project leverages the VADER lexicon, a tool specifically optimized for the emotionally expressive, grievance-driven language characteristic of CFPB narratives. Furthermore, Duru and Güllü (2025) demonstrated the efficacy of Latent Dirichlet Allocation (LDA) in categorizing telecommunications grievances, providing the methodological basis for the unsupervised topic modeling pipeline developed in this study.

Fundamentally, modeling consumer escalation is an exercise in reasoning under uncertainty, which is a core tenet of artificial intelligence. Because the true state of a consumer's frustration cannot be directly observed, it

must be probabilistically inferred from observable actions. Building upon this, this paper introduces a novel "Dual-Signal Framework." Initial empirical evaluations revealed that static classification models alone yield moderate predictive strength, while purely sequential models struggle to anticipate severe, sudden escalations. Therefore, the proposed framework hypothesizes that complaint escalation is driven by two distinct, measurable phenomena: procedural friction and event severity. To capture both dimensions, this project deploys a hybrid approach utilizing an XGBoost static classifier as a signal to measure severity in order to analyze immediate content risk, paired symmetrically with a CategoricalHMM serving as a precise tool to analyze behavioral degradation over time. Ultimately, this study demonstrates how intelligent triage can accurately isolate prolonged procedural loops from sudden, high-cost regulatory escalations.

II. Methods

The methodology of this project was executed across four primary phases: feature engineering, target construction, static classification with temporal validation, and synthetic sequential modeling. The computational pipeline was developed using Python to systematically transform unstructured text into predictive mathematical signals. The methodology can be summarized as shown in Figure-1.

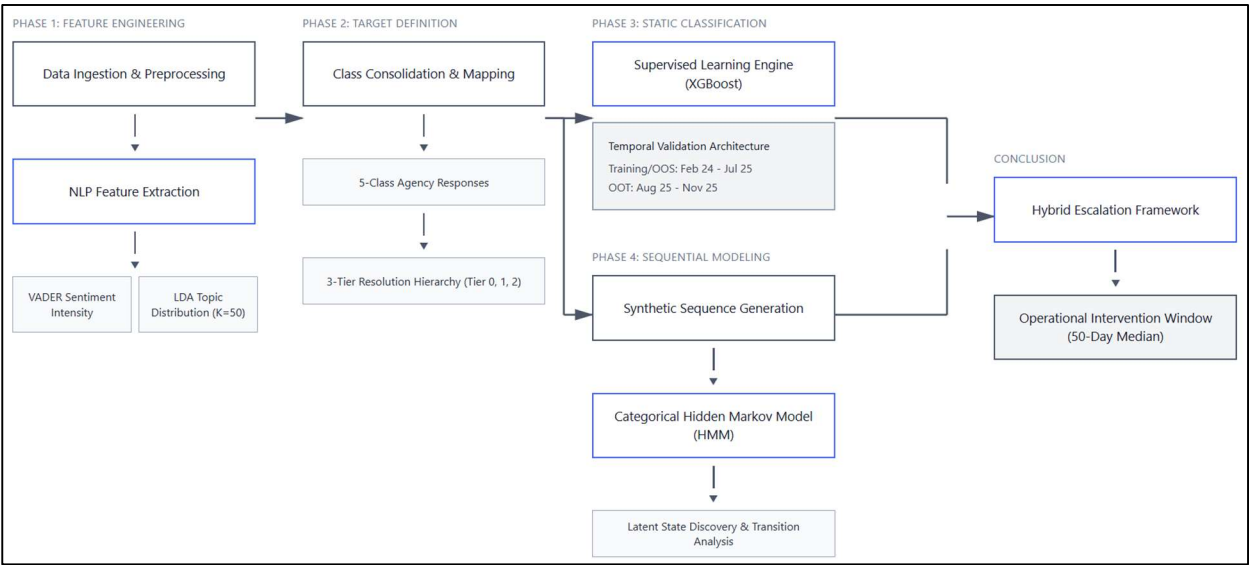


Figure-1: Methodology Workflow

A. Dataset and Tools

I used the Consumer Financial Protection Bureau (CFPB) complaints dataset, which is an openly available dataset to analyze real financial complaints. The website has data of filed consumer complaints from about 15 years. I filtered for the data from February 2024 to February 2026. This resulted in about 1.4M complaint rich data for the period.

The dataset can be downloaded from : <https://www.consumerfinance.gov/data-research/consumer-complaints/#get-the-data>

I also used Chapter-3 of the Speech and Language Processing to evaluate language models and learn about perplexity in N-gram models. It can be viewed here: <https://web.stanford.edu/~jurafsky/slp3/3.pdf>

Additionally, I used Figma to create workflow diagrams.

B. NLP Feature Engineering and Unsupervised Topic Discovery

The first phase involved comprehensive text preprocessing and semantic feature extraction. The raw narrative text of approximately 1.4 million complaints was processed through an automated NLP pipeline encompassing lowercase normalization, special character removal via regular expressions, tokenization, stopword filtration, and lemmatization. Following this normalization, sentiment analysis was conducted. Different NLP techniques were applied on the dataset to evaluate their performance. My initial approach to solve the sentiment analysis problem was using n-gram models (unigrams and bigrams). I used the Bad-of-words technique from scikit-learn to do so. To evaluate these models, I used the perplexity metric. It's a useful metric that is per-word, normalized by length, and provides a way to compare texts of different lengths. Perplexity is a function of probability defined as one over the probability of the data, normalized by the number of words (or tokens). It is calculated after normalizing by the number of words N by taking the N^{th} root. For a test set $W = w_1 w_2 \dots w_N$,

$$\begin{aligned}\text{perplexity}(W) &= P(w_1 w_2 \dots w_N)^{-\frac{1}{N}} \\ &= \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}\end{aligned}$$

The lower the perplexity of the model is, better the model is in language modelling.

Secondly, I compared the usage of VADER (Valence Aware Dictionary Sentiment Reasoning) and modern BERT based pre-trained models like FinBERT and RoBERT-a. This technique was inferred from my literature review where I read about this lexicon-based approach for analyzing sentiments. It is a simple approach that uses a keyword-based dictionary to analyze the sentiments of social media comments. I evaluated the models using on sample size of 10,000 and 50,000 complaints. The modern BERT based models were skewed towards negative sentiment (as the dataset comprised of complaints only, skewness to negative sentiment is expected). It could be acceptable as the models understand the larger context. However, in this project my goal was to identify a spread-out distribution to acquire the means to measure sentiment intensity. Rather than extracting a simple binary polarity, the VADER lexicon was utilized to extract multidimensional compound, positive, negative, and neutral sub-scores, capturing the varying intensities of consumer distress. On a sample size of 50,000 complaints, the Pearson correlation coefficient between VADER and FinBERT was 0.142 and that between VADER and RoBERT-a was 0.190, both showing a weak correlation (statistically, p-value for both was nearly zero, indicating high significance of the comparison as $p < 0.05$).

To capture the thematic context of the grievances, I trained an unsupervised Latent Dirichlet Allocation (LDA). I observed the use of this technique for topic modelling in my literature review. LDA assumes that documents are generated by a mixture of hidden topics, which is highly applicable to complex financial complaints that may blend billing errors with customer service disputes. The optimal number of topics

was rigorously determined by calculating and plotting perplexity scores across various K configurations and observing the Elbow plot. The perplexity metric measures how well a probability model predicts a sample; the model ultimately converged on 50 latent topics modeled using bigrams. In addition to topic and sentiment distributions, custom linguistic markers—such as urgency scores, churn intent, and loyalty markers—were engineered to capture underlying behavioral intent, yielding a final dense dataset of 12 distinct continuous and categorical NLP features per complaint. Figure-2 summarizes the NLP workflow pipeline as discussed in this section.

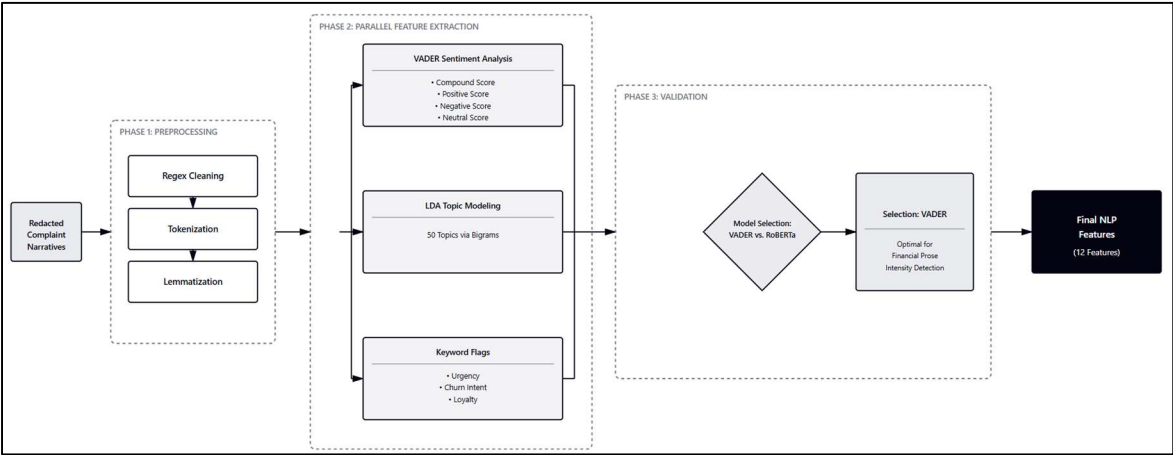


Figure-2: NLP Pipeline

C. Target Construction and Class Consolidation

The second phase required the construction of a mathematically viable target variable. Because the CFPB dataset lacks explicit customer churn labels, the "Company response to consumer" field was analyzed. Initially consisting of five distinct classes, this variable presented significant class imbalance and conceptual overlap. To create a robust target for supervised learning, the responses were consolidated into a three-tier hierarchy representing ascending levels of service complexity and cost. Tier 0 encompasses low-complexity explanations and standard closures, representing 60.4% of the dataset. Tier 1 represents service recovery, defined by non-monetary relief or untimely responses, accounting for 37.8% of the data. Tier 2 denotes high-risk compensation events requiring direct monetary relief, representing a highly skewed minority class of 1.8%. A rigorous substring matching protocol was implemented natively in Python to ensure that strings containing "non-monetary relief" were not erroneously captured by the logic defining "monetary relief," thereby preserving the integrity of the target classes.

D. Supervised Classification and the Generalization Tradeoff

The third phase developed the static engine of the triage framework. To identify immediate content risk which can be defined as what the customer is communicating in the isolated moment of the complaint, several supervised learning classifiers were evaluated. The models tested included Logistic Regression as a linear baseline, Random Forest for ensemble bagging, and XGBoost for gradient boosting. Systematic evaluation was conducted using a strict temporal split rather than a random split to simulate real-world deployment conditions. Data from February 2024 to July 2025 served as the training set, while data from

August 2025 to November 2025 was utilized for out-of-time (OOT) validation. The data from December 2025 to February 2026 was observed to be volatile and thin given the recency of the data.

XGBoost emerged as the superior model, but optimizing it revealed a classic machine learning dilemma regarding the bias-variance tradeoff and temporal generalization. Extensive hyperparameter tuning was conducted using an exhaustive Grid Search strategy, optimizing parameters such as maximum tree depth, learning rate, and subsampling ratios. While the Grid Search successfully identified parameters that improved the Out-Of-Sample (OOS) validation metric on the training timeframe, these optimized parameters caused a noticeable degradation in performance when applied to the Out-Of-Time (OOT) dataset. This divergence indicated that the hyperparameter tuning was overfitting to the specific temporal distribution of the training data. Recognizing the importance of robustness against concept drift in real environments, I decided to discard the grid-searched parameters for time being and proceed with the initial model. The baseline XGBoost configuration was retained, strategically prioritizing future temporal generalization over maximum in-sample optimization.

E. Synthetic Sequence Generation and HMM Architecture

The final methodological phase developed the "Behavioral Engine" utilizing a Hidden Markov Model. Traditional HMMs require empirically observed sequences; because true customer journeys were unavailable, a simulation-based approach was engineered to generate synthetic multi-complaint sequences. This approach relies on construct validity. Links were hypothesized based on a strict, multidimensional heuristic requiring exact matches on the company name, geographic state, LDA topic category, and three-digit ZIP code prefix. Furthermore, temporal constraints required complaints to occur within a 7-to-60-day window, and behavioral logic dictated that sentiment must be the same or worsening to constitute a valid escalation chain. These were the assumptions made to link the existing complaints in the dataset for simulating complaints from same customers. The algorithm was executed on data sorted by date of complaint to respect the original timeline of possible escalation. Once a set of matching sequences was identified, it was assigned a sequence_id which was a proxy for unique customer identifier. The length of these proxy sequences was limited to a chain of 4 complaints. As the approach to simulate the sequences involved a heuristic and no true matching criteria, I decided to stick with a plausible length of Markov chain which would support my exploratory analysis of using HMM on this data in combination with NLP features. This approach resulted in processing 965,226 complaints into 623,939 synthetic sequences. The distribution of different sequence chain length is depicted in Figure-3.

Sno	Sequence Length	Count	% of Total
1	1	443,543	71.09%
2	2	83,146	13.33%
3	3	33,609	5.39%
4	4	63,641	10.20%

Figure-3: Sequence Length Distribution of Complaints

To simplify and model the hidden states and transitions, only multi-length sequences (sequence length>1) were considered as an input to the HMM. The resulting sequences were formalized into an observation matrix for the HMM. Because the observations were discrete states (a matrix of Tier x Sentiment), the architecture was transitioned from a standard Gaussian distribution to a CategoricalHMM. The fundamental assumption of the HMM is the Markov property—that the probability of transitioning to a future state depends entirely on the current state, independent of the preceding history. Training an HMM relies on the Expectation-Maximization (EM) algorithm, which is highly susceptible to converging on local optima rather than the global optimum. To mitigate this, I experimented with ten random restarts instantiation across increased iterations. Model complexity was penalized and optimized using the Bayesian Information Criterion (BIC), which mathematically balances models against the number of parameters. The BIC curve indicated that a robust 8-state architecture (K=8) provided the optimal representation of latent customer frustration levels without overfitting the synthetic sequences.

Furthermore, the number of hidden states were assumed to be 9. Through a combination of predicted resolution tier and sentiment of the complaint sequence, with the sentiment of latest complaint in the sequence used for modelling, 9 hidden states were defined for the model to train upon.

III. Results

The systematic evaluation of hybrid architecture yielded counter-intuitive but highly actionable findings regarding the nature of consumer escalation. Rather than functioning as a single, unified predictive pipeline, the static NLP classifier and the dynamic HMM demonstrated highly specialized, complementary capabilities, validating the Dual-Signal Framework.

A. Exploratory Data Analysis

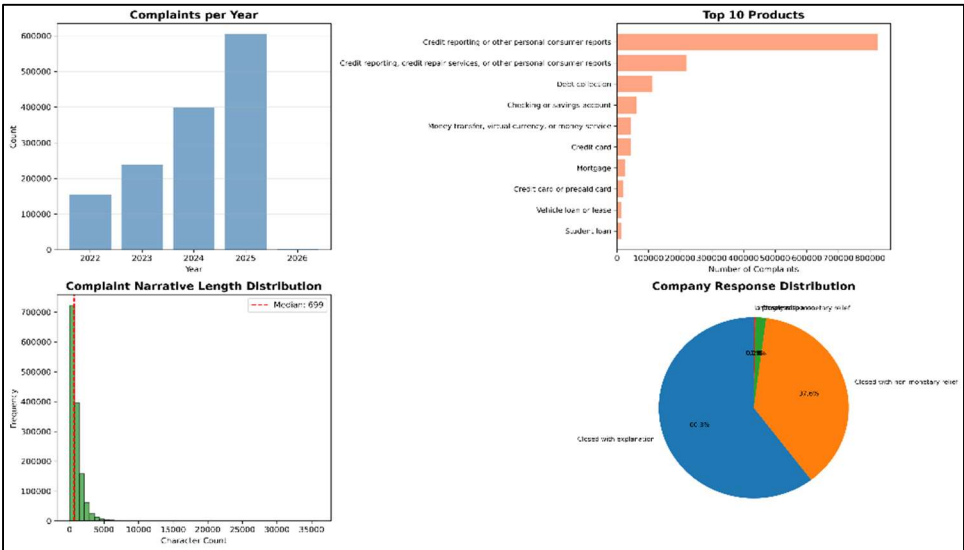


Figure-4: EDA of Product Types and Complaints

Initial exploratory data analysis of the CFPB dataset revealed that the financial complaints per year were rising steadily. A Pareto-style product distribution highlighted that the dataset is heavily dominated by

credit reporting disputes, necessitating the use of Latent Dirichlet Allocation (LDA) to extract distinct thematic context from a skewed feature space. Furthermore, the rising complaints per year necessitate a need for a transparent, efficient and explainable system to process such customer complaints as the economy moves towards consumer-centric businesses. This is shown in Figure-4.

B. Evaluation of NLP Techniques

As discussed in the methodology earlier, the dataset was evaluated on a sample size of 50,000 random complaints for the VADER, FinBERT and RoBERTa models. The sentiment compound in all 3 models can be visualized in Figure-4. It is evident that the VADER technique, although simpler than the modern pre-trained BERT models, works well in the given context of sentiment intensity association as compared to the BERT models which tend to classify majority of the complaints with negative sentiment compound.

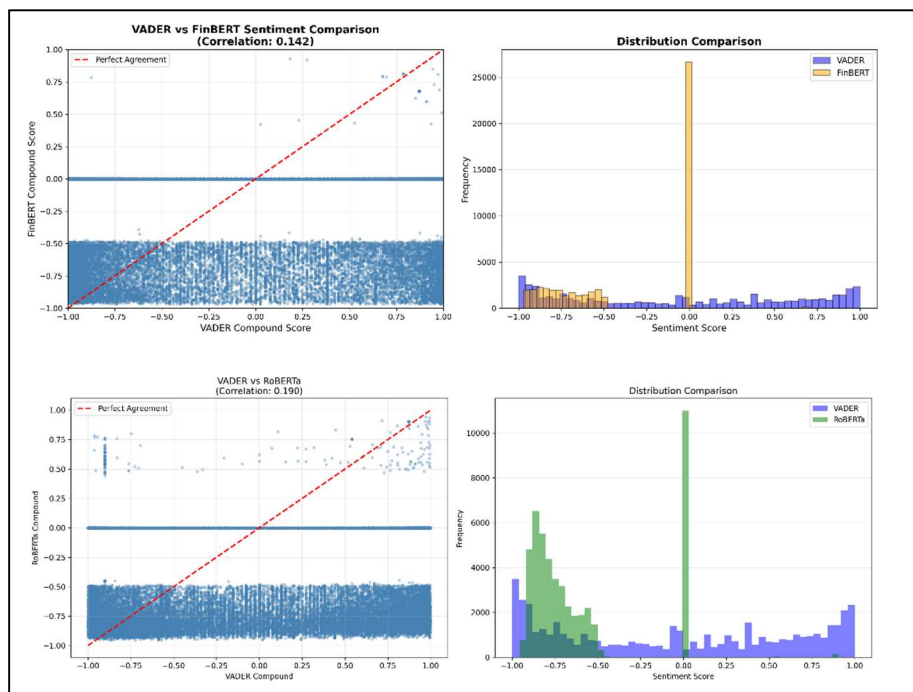


Figure-5: VADER vs Modern BERT models

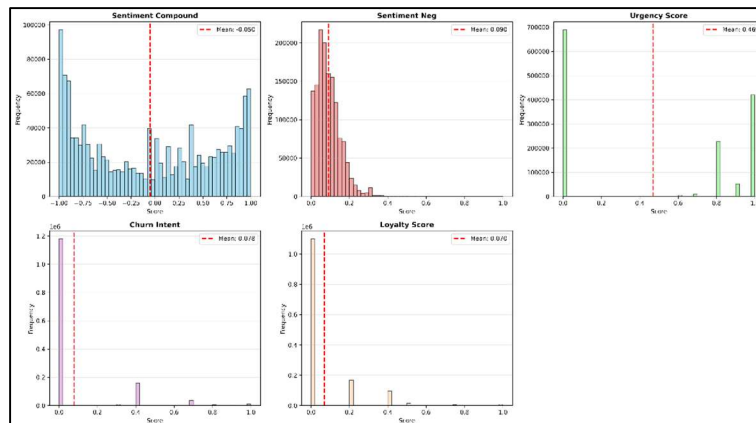


Figure-6: NLP Feature Distribution obtained from VADER

C. Resolution Prediction using XGBoost

The Static Engine, powered by the baseline XGBoost classifier with balanced class weighting, achieved an Out-of-Time Area Under the Curve (OOT AUC) of 0.6775. While this metric denotes moderate standalone predictive power, feature importance analysis revealed the model's specific analytical utility. The LDA topic_category emerged as the overwhelmingly dominant predictor, accounting for 43.6% of the model's predictive weight. Since this feature exhibited more importance, I also evaluated how other NLP features stood out when this feature was excluded. Other NLP features like churn intent and sentiments were still significant.

Sno	Model	OOS_AUC	OOS_Accuracy
1	XGBoost	0.757338	0.560781
2	Random Forest	0.754182	0.557746
3	Logistic Regression	0.638687	0.469949

Figure-7: Model comparison on OOS dataset

As observed in Figure-7, the tree-based ensemble methods significantly outperformed the linear baseline. XGBoost achieved the highest Out-of-Sample (OOS) performance with an AUC of 0.757 and an accuracy of 0.561. While Random Forest showed competitive results (AUC 0.754), XGBoost was selected for the final Dual-Signal Framework due to its superior ability to handle the high-dimensional space produced by the LDA topic modeling and VADER sentiment analysis. This effectively acts as a severity filter, evaluating the inherent legal risk embedded within the text of a single grievance, independent of previous interactions.

D. Hidden State Discovery and the Deterministic Loop

The Behavioral Engine provided profound insights into procedural friction through the discovery of eight distinct hidden states. CategoricalHMM successfully segregated the 155,369 multi-complaint sequences into latent behavioral profiles. For example, State H0 was identified as a "Satisfied Resolution" state, capturing instances where sentiment remained highly positive (+0.697) despite the necessity of a Tier 1 service recovery. The HMM matrix for the simulated multi-sequence chains is depicted in Figure-8a.

The most critical finding within the HMM transition matrix was the identification of specific "Escalation Gateways." The model identified a deterministic loop between Hidden State 5 (H5) and Hidden State 7 (H7). Complaints entering this specific behavioral loop exhibited an exceptionally high transition probability of cycling continuously between these high-friction states. This phenomenon allowed for the mapping of discrete risk categories based purely on sequence behavior. States 5 and 7 were categorized as High Risk due to this friction loop; States 0 and 2 were designated as Medium Risk gateways

characterized by escalating negative sentiment; and States 1, 3, 4, and 6 were classified as Low Risk, representing sticky but routine operational interactions. This is shown in Figure-8b.

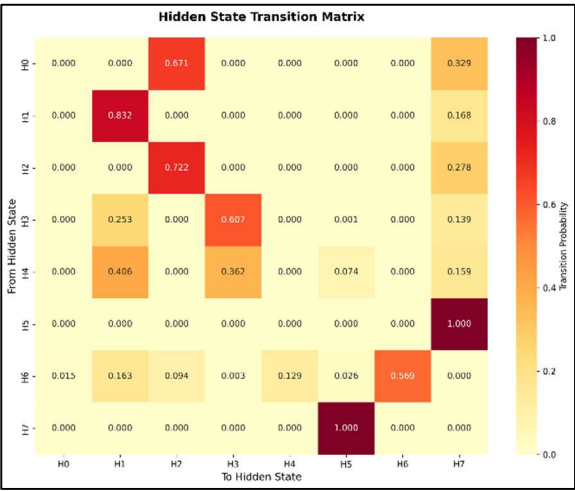


Figure-8a: HMM Transition Matrix



Figure-8b: Emission in hidden states and risk categorization

E. Dual-Signal Framework

The integration of these modeled risk states against the final resolution tiers revealed important outcomes regarding behavioral modelling on this dataset. A row-wise normalization of predictive precision demonstrated that the HMM is an extraordinarily powerful predictor of Tier 1 service recovery. Complaints exhibiting High or Medium behavioral risk possessed an approximate 50% probability of concluding in a Tier 1 resolution, compared to only 35% for Low-risk sequences. The sequential model successfully captures most of all Tier 1 resolutions, proving its efficacy in identifying customers trapped in procedural friction.

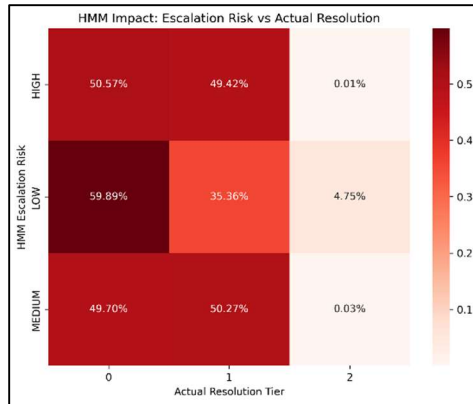


Figure-9: Impact of HMM on resolution tiers showing escalation risk vs actual risk

Furthermore, temporal pathway analysis of the overall 21.5% escalation rate revealed a median timeframe of 50.0 days for a complaint to escalate. This can be observed from Figure-10. This finding translates to a critical seven-week intervention window wherein organizations can apply targeted service recovery to disrupt the H5-H7 deterministic loop before systemic failure occurs.

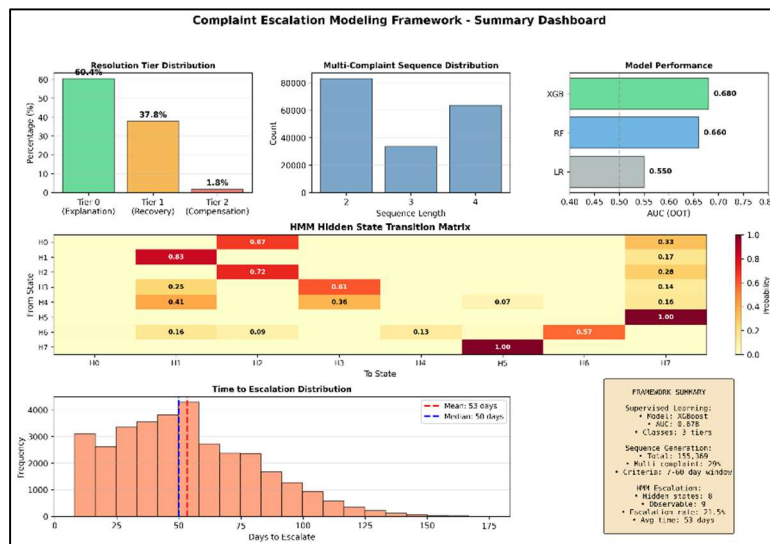


Figure-10: Stats obtained after customer escalation modelling

IV. Conclusions

The findings of this project validate the strategic necessity of the proposed Dual-Signal Framework, while advancing the objective of Explainable AI (XAI) within financial services. The overarching conclusion is that a single modeling methodology—whether entirely static or entirely sequential—is insufficient to capture the dual nature of consumer escalation in complex financial services.

The strategic business value of this framework lies in the specialization of its components. The HMM provides temporal interpretability; instead of a black-box churn score, it identifies a customer's specific movement through latent frustration states, highlighting the "deterministic loops" (States 5 ↔ 7) that

necessitate human intervention. Conversely, the XGBoost classifier provides semantic interpretability through feature importance. By identifying that 43.6% of predictive weight rests on specific topic categories, the model allows compliance officers to understand the exact regulatory triggers driving high-risk classifications.

The primary limitation of this project is its reliance on synthetically generated sequences. While the strict linking heuristics provide strong construct validity—evidenced by the logical sentiment degradation and thematic consistency—the sequences are simulated hypotheses rather than empirically observed truths. Furthermore, the detection of severe temporal distribution drift in data from December 2025 to January 2026, where Tier 0 responses spiked to 85% ($p < 0.0001$), indicates that consumer reporting patterns are highly sensitive to macro-environmental shifts. This requires any deployment of this system to incorporate continuous monitoring and periodic retraining.

Since the methodology involved usage of simulated sequences for HMM, there were many assumptions made to experiment with the behavioral modeling aspect of consumer complaints to model escalation risk. These assumptions include – heuristic for linking complaints, defining 9 hidden states and using just the sentiment compound feature for the hidden states. The HMM methodology could be improved in the future by using the topics obtained from LDA to define the hidden states given their high feature importance. However, this approach might also result in data becoming thin as there were 50 topics modelled using LDA, which may lead to transition matrix with sticky states.

Future scope and improvements should focus on validating real sequence integrity. If regulatory bodies or private institutions provide actual customer identifiers, the model must be empirically tested to confirm whether real-world Tier 2 cases exhibit more drifting behavior than the simulated sequences currently suggest. Additionally, future technical iterations should operationalize this framework by developing a weighted hybrid scoring equation. By calculating a unified metric, such as $\text{Final Risk} = (\alpha \times \text{NLP Probability}) + (\beta \times \text{HMM Risk})$, financial organizations could deploy a single intelligent routing architecture that dynamically balances content severity with behavioral friction in real-time.

V. Works Cited

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